Theory of Fixed-Axis Trajectory Constraint Classification

1 Problem Statement

Given a discrete trajectory

$$p_k \in \mathbb{R}^3, \qquad k = 1, \dots, N,$$

the goal is to determine whether the motion of a single tracked point is best explained by one of the following fixed-axis motion models:

- circular motion around an axis,
- translation parallel to an axis,
- screw motion around an axis.

The method is formulated directly in discrete time from sampled positions. It does not rely on instantaneous velocities or continuous-time kinematics.

2 Fixed Axis Representation

A candidate axis is represented as

$$\mathcal{L} = \{c + \lambda u : \lambda \in \mathbb{R}\}, \qquad \|u\| = 1,$$

where:

- $u \in \mathbb{R}^3$ is the axis direction,
- $c \in \mathbb{R}^3$ is a canonical point on the axis.

The implementation uses the gauge condition

$$u^\top c = 0,$$

which makes c the point on the axis closest to the origin.

3 Cylinder-Like Common Geometric Model

All three motion classes are naturally associated with a fixed axis and an approximately constant radial offset. A common fitting objective is therefore

$$E_{\mathrm{cyl}}(u, c, r) = \sum_{k=1}^{N} \left(\left\| (I - uu^\top)(p_k - c) \right\| - r \right)^2,$$

where r is the distance from the trajectory to the axis.

Here,

$$(I - uu^\top)(p_k - c)$$

is the component of the point displacement orthogonal to the axis direction u.

In the implementation, this objective is minimized numerically with nonlinear least squares.

4 Axis-Aligned Orthonormal Frame

Once an axis direction u is known, an orthonormal frame (e_1, e_2, u) is constructed such that

$$e_1^\top u = 0, \qquad e_2^\top u = 0, \qquad e_2 = u \times e_1.$$

Define

$$B = [e_1\ e_2\ u].$$

This matrix maps the trajectory into a coordinate frame aligned with the candidate axis.

5 Cylindrical Coordinates

For each sample p_k, define

$$x_k = B^\top (p_k - c), \qquad x_k = \begin{bmatrix} x_k^{(1)} \\ x_k^{(2)} \\ x_k^{(3)} \end{bmatrix}.$$

The cylindrical coordinates are then

$$\rho_k = \sqrt{(x_k^{(1)})^2 + (x_k^{(2)})^2},$$

$$\phi_k = \operatorname{atan2}\left(x_k^{(2)}, x_k^{(1)}\right),$$

$$h_k = x_k^{(3)}.$$

The angular sequence ϕ_k is unwrapped over sample order.

6 Motion Models

6.1 Circular Motion

For ideal circular motion around the axis,

$$\rho_k \approx r, \qquad h_k \approx h_0,$$

while the angle ϕ_k varies.

The residual is

$$E_{\text{circ}} = \sum_{k=1}^{N} \left[(\rho_k - r)^2 + (h_k - h_0)^2\right].$$

In the implementation:

- r is estimated as the mean of ρ_k,
- h_0 is estimated as the mean of h_k.

The fitted three-dimensional circle center is

$$p_{\text{center}} = c + h_0 u.$$

This is not the same as the reported axis point c.

6.2 Translation Parallel to the Axis

For ideal translation parallel to the axis,

$$\rho_k \approx r, \qquad \phi_k \approx \phi_0,$$

while the axial coordinate h_k varies.

The residual is

$$E_{\text{trans}} = \sum_{k=1}^{N} \left[(\rho_k - r)^2 + d_\angle(\phi_k, \phi_0)^2 \right],$$

where $d_\angle$ denotes wrapped angular distance.

An important identifiability issue appears here: for a single tracked point undergoing pure translation, the direction of the axis is observable, but the exact parallel axis location is not unique. Therefore, the translation model should be interpreted mainly through axis direction and displacement, not through a unique recovered axis point.

6.3 Screw Motion

For ideal screw motion,

$$\rho_k \approx r, \qquad h_k \approx h_0 + \kappa \phi_k,$$

where κ is the pitch per radian.

The residual is

$$E_{\text{screw}} = \sum_{k=1}^{N} \left[(\rho_k - r)^2 + (h_k - h_0 - \kappa \phi_k)^2 \right].$$

In the implementation:

- r is estimated as the mean of ρ_k,
- (h_0, κ) are estimated by linear least squares.

7 Practical Estimation Strategy

The implementation uses a pragmatic multi-hypothesis strategy rather than a single monolithic global optimization.

7.1 Circle Hypothesis

The axis direction is initialized from the PCA direction with the smallest variance. This is motivated by the fact that a circular trajectory approximately lies in a plane, so the axis is close to the normal of that plane.

7.2 Translation Hypothesis

The axis direction is initialized from the PCA direction with the largest variance, since translation of a single point produces a dominant line direction.

7.3 Screw Hypothesis

For screw motion, a general nonlinear cylinder fit is used to estimate axis direction, axis point, and radius.

8 Classification

The three model residuals

$$E_{\text{circ}}, \qquad E_{\text{trans}}, \qquad E_{\text{screw}}$$

are computed and compared. The predicted class is the model with the smallest residual.

The code also reports:

- normalized residuals,
- a confidence score based on the gap between the best and second-best residuals,
- motion-extent summaries from the fitted cylindrical coordinates.

9 Estimated Motion Extents

After selecting the best-fitting model, the fitted cylindrical coordinates are used to summarize the motion.

9.1 Circle

For circular motion, the code reports:

- estimated start angle,
- estimated end angle,
- estimated angular span.

9.2 Translation

For translation, the code reports:

- axial displacement.

9.3 Screw

For screw motion, the code reports:

- angular span,
- axial displacement.

10 Limitations

- Short arcs can be ambiguous and may resemble line segments or short helix fragments.
- Very small radius values lead to ill-conditioned axis estimation.
- For pure translation of a single tracked point, the parallel axis position is not uniquely identifiable.
- The method is a readable prototype designed for clarity rather than maximum estimation sophistication.

11 Relation to the Implementation

The theory described here is implemented in:

- `trajectory_constraint_fit.py`,
- `simulate_trajectories.py`,
- `demo.py`.